DATA FACTSHEET

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Digital Trace Data: Experimental

Non-traditional indicators for humanitarian response

COMPLEMENTS TRADITIONAL DISPLACEMENT DATA

Digital trace data are passively generated records of human activity — the GPS signals from mobile phones, web and search activity, satellite earth observation, and platform-derived population estimates. Unlike traditional displacement figures compiled through field surveys, registration or enumeration (IOM's **Displacement Tracking Matrix**, **UNHCR** and **IDMC**), these sources are produced continuously and can, in some cases, be observed in near real time. This factsheet summarises the main indicators, what each can and cannot tell us, and how they **complement — rather than replace** — authoritative humanitarian reporting.

Why it matters for response

- **Timeliness.** Signals available within hours to days, ahead of much field reporting.
- **Granularity.** Sub-district patterns of movement, not just headline totals.
- **Gap-filling.** Coverage where access is constrained or warning systems are absent.
- **Triangulation.** Cross-checks and contextualises official figures.

Mobility indicators

GPS / mobile-device mobility data

Used primarily for: near-real-time population change, displacement and return.

Device-level GPS locations from mobile phones, sourced from a commercial vendor that **aggregates many apps** to reduce single-source bias. Device counts are scaled to population using **WorldPop**. Movement is measured as the **deviation from a modelled baseline** of usual mobility, so the change attributable to an event can be isolated from normal day-to-day variation. Flooded administrative units are compared against unflooded ones in a **difference-in-differences** design, with a displacement threshold (typically one day, with sensitivity checks at three and seven days). Its strengths are **timeliness** and **spatial granularity**, down to sub-district level.

Considerations
- Not a representative sample — excludes people without a smartphone or data plan.
- Requires baseline modelling and a control group; figures are indicative of patterns, not precise counts.
- Commercial access and cost; data are aggregated, with no individual identification.

Web & search activity

Used primarily for: attention and information-seeking signals.

- **Google search activity** (Trends): relative interest in terms such as *banjir* (flood) or evacuation — an early signal of public concern.
- **Wikipedia pageviews:** spikes in views of event-related articles as an attention proxy.
- **Cloudflare HTTPS requests:** regional internet traffic that can flag connectivity disruption during a disaster.

Considerations
- Attention is not displacement; strongly confounded by media coverage.
- Coarse spatial resolution and language-dependent.

Meta (Facebook) population & movement

Used primarily for: crisis population baselines and movement flows.

Released through Meta's **Data for Good** programme in the aftermath of major humanitarian disasters. Once posted, the data are available to researchers and policymakers for **90 days** before removal. They show the number of Facebook users in a spatial unit at a given time.

- **Population (stock):** users per unit at three daily snapshots — 00:00, 08:00 and 16:00 (Pacific Time).
- **Movement (flow):** origin–destination of users between snapshots, based on where each user spent most time in each 8-hour window.
- **Scales:** 800 m tiles (Bing Maps quadkeys) and GADM level-2 administrative units.

Considerations
- Cells or flows with fewer than **10 observations** are suppressed.
- Coverage gaps occur — e.g. Pakistan Movement data only at 08:00 and 16:00, some Population dates missing, and the 2025 Karachi event available only at tile level.
- Time-limited (90-day) availability; Facebook users are not representative of the wider population.

Contextual indicators

Earth observation

Used primarily for: physical context — population baselines, rainfall and flood extent.

- **Population:** WorldPop gridded population estimates, used to scale and contextualise figures.
- **Rainfall:** NASA Global Precipitation Measurement (GPM) IMERG Late Run — near-real-time precipitation to locate heavy-rainfall areas.
- **Flood:** NASA Land, Atmosphere Near real-time Capability for Earth observation (LANCE), MODIS satellite — near-real-time flood extent.

Considerations

- Measures the hazard and physical setting, not people or displacement directly.
- Cloud cover and spatial resolution constrain operational use.

Cross-cutting considerations

- **Representativeness.** Every source carries coverage bias; figures are indicative of patterns, not exact counts.
- **Validation.** Estimates should be cross-checked against authoritative sources — IOM DTM, UNHCR, IDMC, OCHA and national authorities.
- **Privacy & ethics.** Data are aggregated with no individual identification, and governed by data-sharing agreements.
- **Complement, not replacement.** Digital traces are strongest when triangulated with traditional reporting, not used in isolation.
- **Access & sustainability.** Several sources are commercial or platform-controlled and may change without notice.

Data source	Access	Main limitations
GPS / mobile-device mobility	Commercial — licensing can be expensive	Smartphone users only (not representative); needs a modelled baseline and control group
Web & search activity (Google Trends, Wikipedia)	Free and public	Signals attention, not displacement; coarse geography; confounded by media coverage
Cloudflare HTTPS requests	Free (Cloudflare Radar)	Connectivity proxy only; aggregate, with no demographic detail
Earth observation (WorldPop, GPM rainfall, MODIS flood)	Free and open	Measures the hazard and physical setting, not people; cloud cover and resolution limit use
Meta Data for Good (population & movement)	Free, but released only for selected crises; 90-day window	Facebook users only; small counts suppressed; coverage gaps between contexts

Table 1. Indicative summary of digital trace data sources, how they are accessed, and their principal limitations. Suitability depends on the context, the event and the data available at the time.

DRAFT — outstanding to-dos (not for circulation)

- Add hyperlinks to each source — UNHCR Refugee Data Finder, IOM DTM, IDMC, WorldPop, NASA GPM IMERG and LANCE/MODIS, Cloudflare Radar, Google Trends, Meta Data for Good.
- Expand the description and explanation of each data source — what it measures, spatial resolution and update frequency.
- Add a short “how to interpret” note for policymakers.
- Add a worked example linking to a published SitRep (e.g. Indonesia, Iran).
- Confirm cost and licensing detail for the commercial GPS vendor.
- Add a brief glossary (quadkey, GADM, difference-in-differences).
- Add references and a methodology link.
- Decide whether to reinstate GloFAS and night-time lights.
- Add version number, date and contact / authorship.